POOJA B E

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PROFESSIONAL SUMMARY

3rd-semester Electronics and Communication Engineering student with a growing focus on VLSI and digital design. Hands-on exposure to Verilog RTL design, digital logic, Arduino-based electronics, PCB design using KiCad, circuit simulation, and basic embedded systems. Interested in RTL design, digital hardware, verification, and practical electronics projects.

TECHNICAL SKILLS

- VLSI / Digital Design: Verilog HDL, combinational & sequential logic, counters, multiplexers/demultiplexers, comparators, RTL simulation, testbenches
- Electronics: Digital Electronics, Electronic Devices, Circuit Theory, semiconductor fundamentals, basic circuit analysis
- Embedded / Arduino: Arduino, IR sensor modules, buzzer/audio interfacing, basic Embedded C, breadboard prototyping
- PCB / Simulation Tools: KiCad schematic capture, PCB layout/routing, DRC concepts, Wokwi circuit simulation

PROJECTS

- **Verilog RTL & Digital Design Simulations:-** Designed and simulated digital modules including full adder, 4-bit magnitude comparator, counters, 8:1 multiplexer, 4:1 demultiplexer, and BCD-to-seven-segment decoder; created testbenches for functional verification.
- **IR Arduino Sensor Module PCB Design:-** Designed an Arduino-compatible IR sensor module PCB, working through schematic planning, component connections, PCB layout and routing, and basic design-rule considerations.
- **Arduino Retro Music / 8-bit Music Project:-** Built an Arduino-based retro music project using a buzzer to generate programmed musical tones and melodies, gaining practical experience with Arduino coding, timing, note frequencies, and hardware interfacing.
- **24V-to-5V Buck / Regulator PCB:-** Worked on component selection and PCB design considerations for a 12V-to-5V power-conversion board, including resistor, capacitor, inductor and diode selection, schematic planning and PCB layout.
- **NE555P Timer PCB Design:-** Designed and developed a PCB based on the NE555P timer IC, including schematic design, component selection, PCB layout, routing, and basic design-rule considerations.
- **Medi Safe AI — Healthcare Technology Project:-** Independently developed/presented a healthcare-focused technology project exploring AI for real-world healthcare challenges. Selected in the Top 50 among 4,676 participating teams in the Bharat Academix Global Innovation Hackathon 2026.

ACHIEVEMENT

- Top 50 / 4,676 teams — Bharat Academix Global Innovation Hackathon 2026; individual participation with Medi Safe AI.

AREA OF INTEREST

VLSI Design • RTL Design • Digital Verification • FPGA • Embedded Systems • PCB Design